

# Day 02 - Introduction to Generative AI and Agents

**Platform:** Microsoft Learn

**Status:** Completed

## Learning Objectives

Understand Generative AI, LLMs, tokenization, embeddings, attention mechanism, system prompts, AI agents, and their cybersecurity applications.

## What is Generative AI?

Generative AI creates new content such as text, images, code, audio, and video. Examples include ChatGPT, Microsoft Copilot, Google Gemini, Claude, and DALL-E.

## Large Language Models (LLMs)

LLMs are trained on massive text datasets and can understand and generate human-like language.

## Tokenization

Tokenization splits text into smaller units called tokens so AI models can process language.

## Embeddings

Embeddings are vector representations that capture the semantic meaning of words and sentences.

## Attention Mechanism

Attention helps the model focus on the most relevant words when understanding context.

## System Prompt

A system prompt defines the AI assistant's role, behavior, and rules.

## AI Agents

AI agents perform tasks, make decisions, and use tools to achieve goals.

## Cybersecurity Applications

- Incident report writing
- Log analysis
- Phishing investigation
- Malware summaries
- Threat intelligence summaries

## Key Takeaways

Always verify AI-generated content before using it in real-world security decisions.

## Interview Questions

1. What is Generative AI?
2. What is an LLM?

3. What is Tokenization?
4. What are Embeddings?
5. What is a System Prompt?
6. What is an AI Agent?